

Deep Learning Model Used- Bidirectional Long Short-Term Memory (LSTM)

A Bidirectional Long Short-Term Memory (BiLSTM) network is a type of Recurrent Neural Network (RNN) that processes sequence data in both forward and backward directions to capture past and future context simultaneously.

1. Why Bidirectional LSTM Model

The project requires analyzing textual reviews and classifying them into discrete rating categories (1–5).

A Bidirectional Long Short-Term Memory (BiLSTM) model was chosen because:

- LSTM units can capture long-term dependencies in text, making them effective for understanding the context of a sentence.

- Bidirectional processing allows the model to learn from both past (previous words) and future (upcoming words) context in the text.

- This dual-direction approach improves the ability to interpret nuanced language, which is critical for accurate rating prediction.

2. Model Architecture Overview

1. **Embedding Layer**

- Input: Tokenized and padded sequences.

- Function: Converts integer-encoded words into dense vector representations of size 128.

- Purpose: Captures semantic meaning of words in a continuous vector space.

2. **Bidirectional LSTM Layer (64 units)**

- Processes the sequence forward and backward, returning sequences for the next recurrent layer.

3. **Bidirectional LSTM Layer (32 units)**

- Further processes the output to capture deeper contextual relationships.

4. **Dense Layer (64 units, ReLU activation)**

- Adds non-linearity and enables learning complex relationships.

5. **Dropout Layer (0.5)**

-Randomly drops 50% of neurons during training to reduce overfitting.

6. **Dense Output Layer (Softmax activation)**

-Output size: num_labels (number of rating classes).

-Produces a probability distribution across all rating categories.

3.Workflow in LSTM Model

1.Input:

Preprocessed text (cleaning, tokenization, and padding) enters the embedding layer.

2.Feature Extraction:

The Bidirectional LSTMs capture contextual information in both forward and backward directions.

3.Representation Learning:

Dense layers transform learned sequences into class-specific features.

4.Prediction:

The softmax output layer assigns a probability to each rating category.

5Decision:

The rating with the highest probability is selected as the prediction.

4.Training Details

-**Optimizer:** Adam (adaptive learning rate optimization).

-**Loss Function:** Categorical Crossentropy (multi-class classification).

-**Batch Size:** 32.

-**Epochs:** Up to 20, with Early Stopping after 3 epochs without improvement in validation loss.

-**Class Weighting:** Applied to handle class imbalance.

-**Validation:** Performance monitored on a held-out validation dataset during training.

5. Advantages

- Context-Aware:** Learns meaning from both past and future words.
- Handles Long Sequences:** LSTM cells mitigate vanishing gradient issues common in standard RNNs.
- Improved Performance:** Generally outperforms unidirectional LSTM in sentiment and rating tasks.
- Flexibility:** Works with variable-length input sequences after padding.

6. Limitations

- Computationally Expensive:** Requires more memory and training time than simpler models.
- Overfitting Risk:** Needs dropout and regularization, especially with small datasets.
- Performance on Mid-Range Ratings:** Struggles to differentiate between close categories (e.g., ratings 2–4) due to similar language patterns.
- Interpretability:** Harder to explain model decisions compared to simpler statistical methods.